

Network Routing Simulation – Revised Answer Key

Explains why each answer is correct based on the routing tables shown in the PBQ.

Scenario: Users cannot access files on File Server 2. The task is to validate routing between Workstation A and File Server 2.

Router	Problem Found?	Required Answer	Why this is correct
Router A	No	No static route required	Router A already has the Site A networks directly connected, including 10.0.4.0/22, and it also has a default route out GigabitEthernet1 toward the rest of the network. Nothing in its routing table prevents traffic from leaving Site A.
Router B	Yes	Add 10.0.4.0/22 via GigabitEthernet1	Router B only knows its directly connected networks: 10.0.0.0/22 and 172.16.27.4/30. It does not have a route to 10.0.4.0/22, which is where File Server 2 lives, and it also does not have a default route. Because GigabitEthernet1 points toward Router C, adding a static route for 10.0.4.0 with mask 255.255.252.0 out GigabitEthernet1 correctly sends traffic toward Site A.
Router C	No	No static route required	Router C already has the needed routes for both sides of the path: 10.0.0.0/22 via GigabitEthernet1 and 10.0.4.0/22 via GigabitEthernet2. It also has both transit links directly connected, so it is already routing traffic between Router B and Router A correctly.

Key takeaway: The only routing gap is on Router B. The missing route to 10.0.4.0/22 prevents Workstation A's side from reaching File Server 2's network through Router C.

Answer from original key: Router A = No, Router B = Yes (10.0.4.0/22 via GigabitEthernet1), Router C = No.